



**BizFirstAi**

# The Intelligent Enterprise Mesh

Overall Architecture of the BizFirstAi Platform

Synthesizing visual workflows, real-time messaging, and agentic AI into a singular operating system.



# Platform Executive Synthesis

A unified enterprise-grade ecosystem eliminating the friction between form management, real-time event orchestration, and AI agent reasoning.

17

Integrated Modules



90+

Total Node Types



605+

Database Tables



<100ms

Messaging Latency





# Diagnostic Matrix: Fragmented Stack vs. Unified Architecture

| Architectural Layer     | The Fragmented Status Quo          | The BizFirstAi Unified Mesh                         |
|-------------------------|------------------------------------|-----------------------------------------------------|
| Forms/UI                | Custom React & Legacy Integrations | Atlas FormStudio<br>(89+ schema-driven controls)    |
| Workflows               | Zapier / MuleSoft Patchworks       | FlowStudio<br>(Visual DAG Orchestration)            |
| Security                | Disjointed IAM via Okta            | 4-Layer ProcessSecurity & Hardware TEE              |
| Observability           | \$1M+ Datadog Overlays             | Built-in OpenTelemetry<br>(Tempo, Loki, Prometheus) |
| Total Cost of Ownership | \$1.16M+ Annually                  | <b>78%</b> Cost Reduction<br>(\$250k Annually)      |



# The BizFirst Macro-Architecture (The Master Stack)

## Presentation

Atlas Forms, FlowStudio, EdgeInteract

## Nervous System

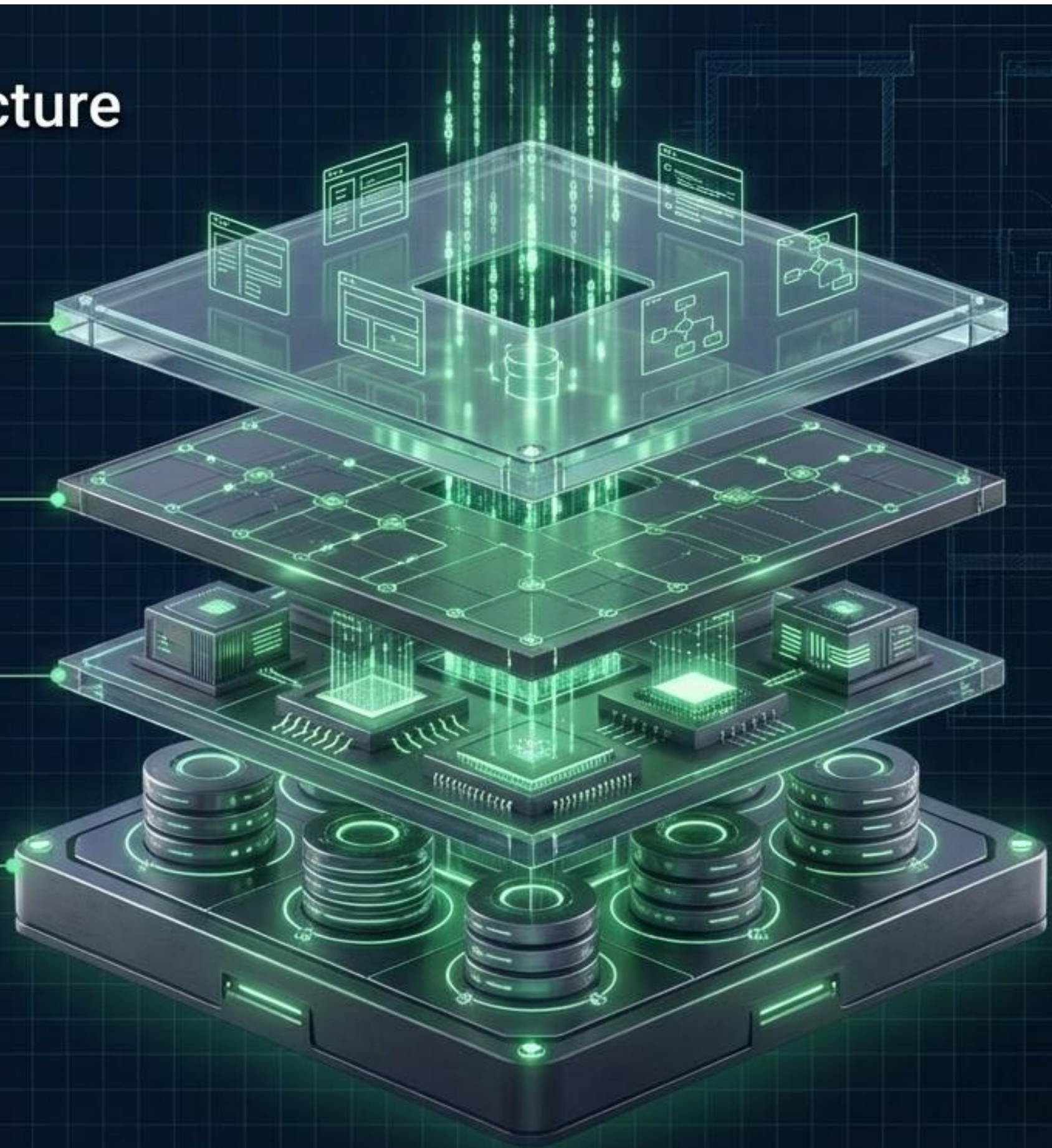
EdgeStream, MeshService

## Brain/Execution

ProcessEngine, NodeHost,  
ANCP, ApprovalNode

## Foundation

DataOcean, ProcessSecurity,  
Passport IAM, Observability



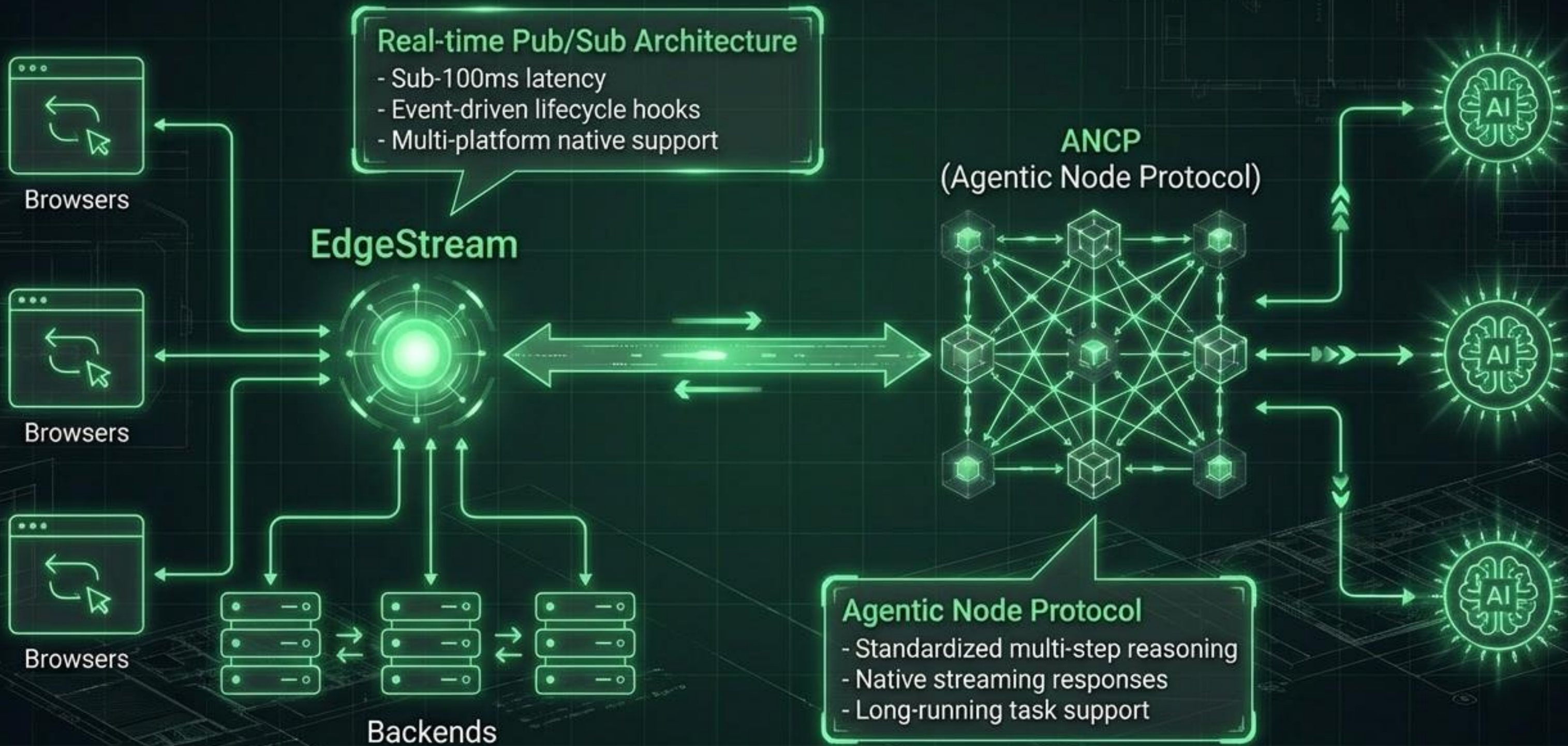


# Presentation Layer: Dynamic UI Orchestration



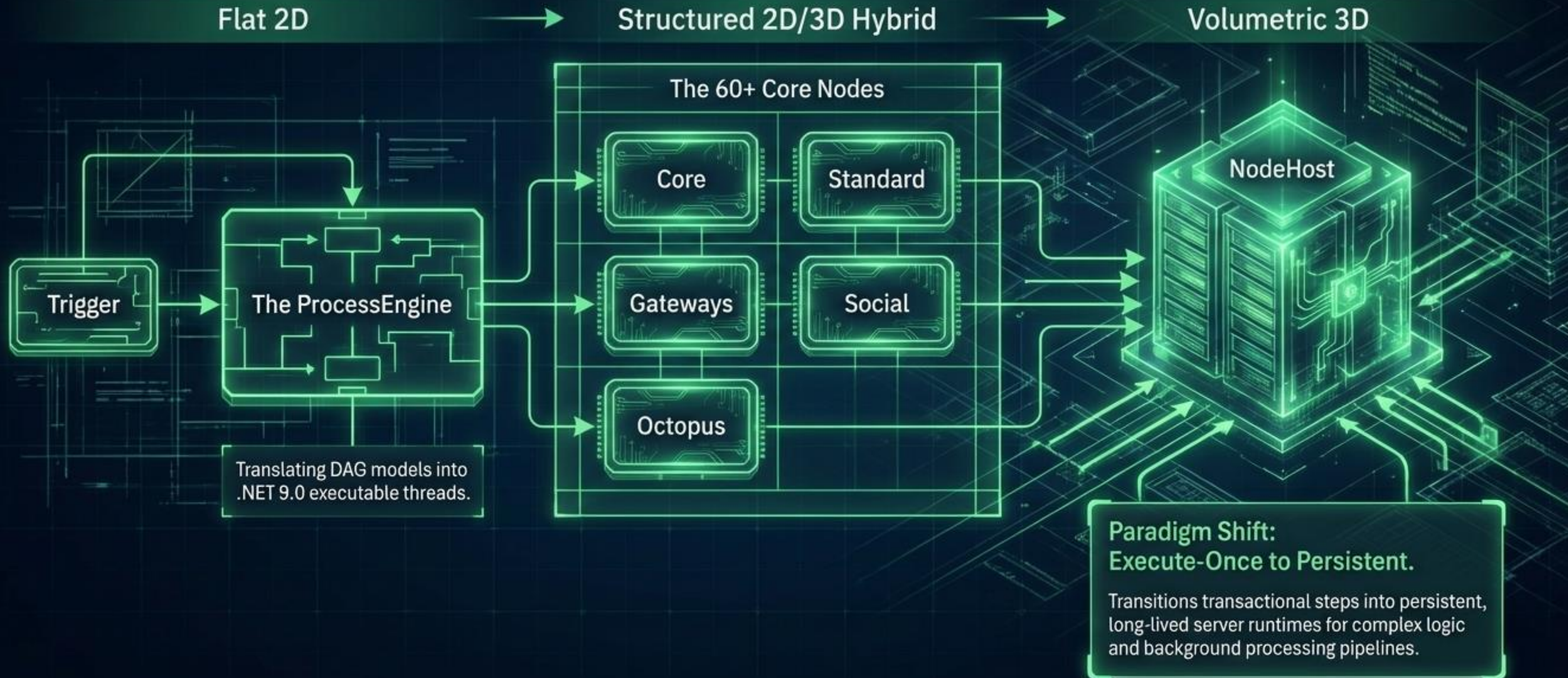


# The Nervous System: EdgeStream & ANCP Mesh



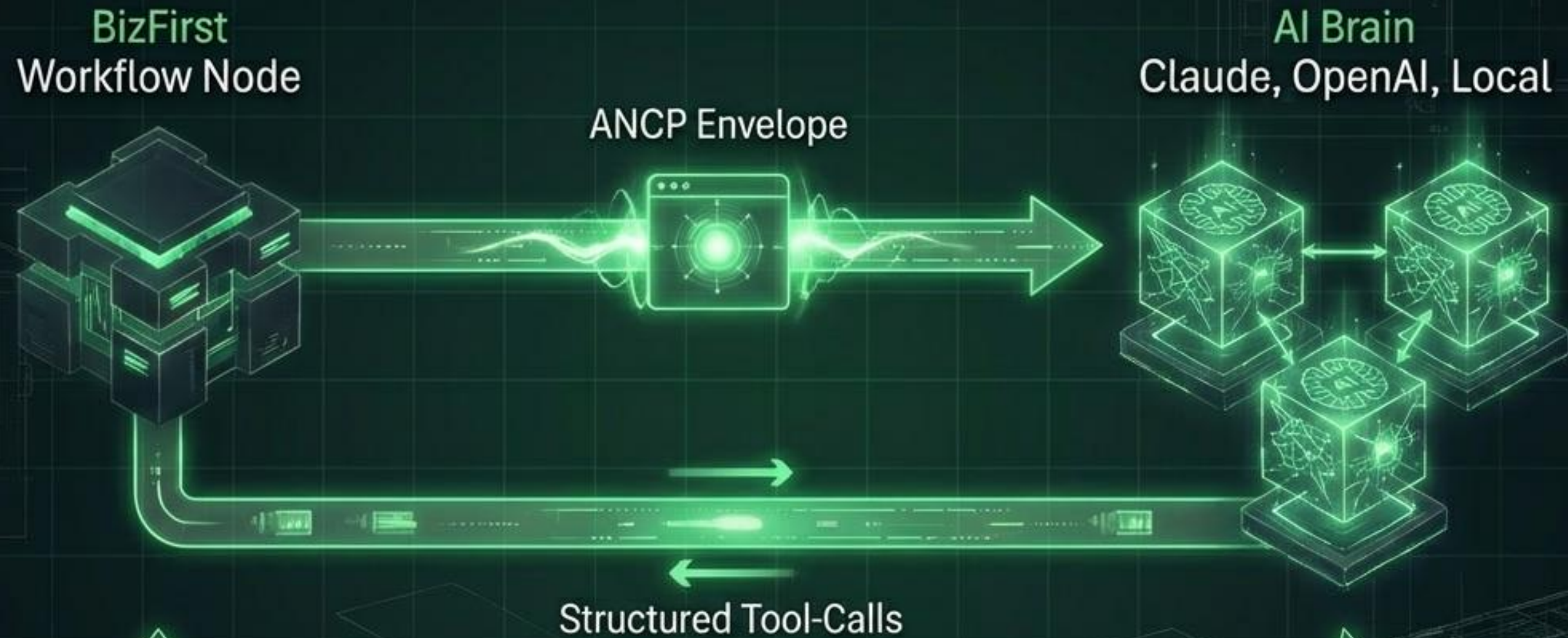


# Execution Layer: NodeHost & ProcessEngine





# The AI Orchestration Engine



## Native AI Orchestration

Multi-step reasoning loops occur directly within the workflow graph, not as bolted-on API calls.

## Agentic Tool Calling

LLMs autonomously request data or trigger external functions via secure ANCP routing before returning to the main execution thread.



# Human-In-The-Loop (ApprovalNode Architecture)

**Phase 1: Atomic Suspension**  
Workflow pauses securely. SignalR pushes real-time notifications to actors.

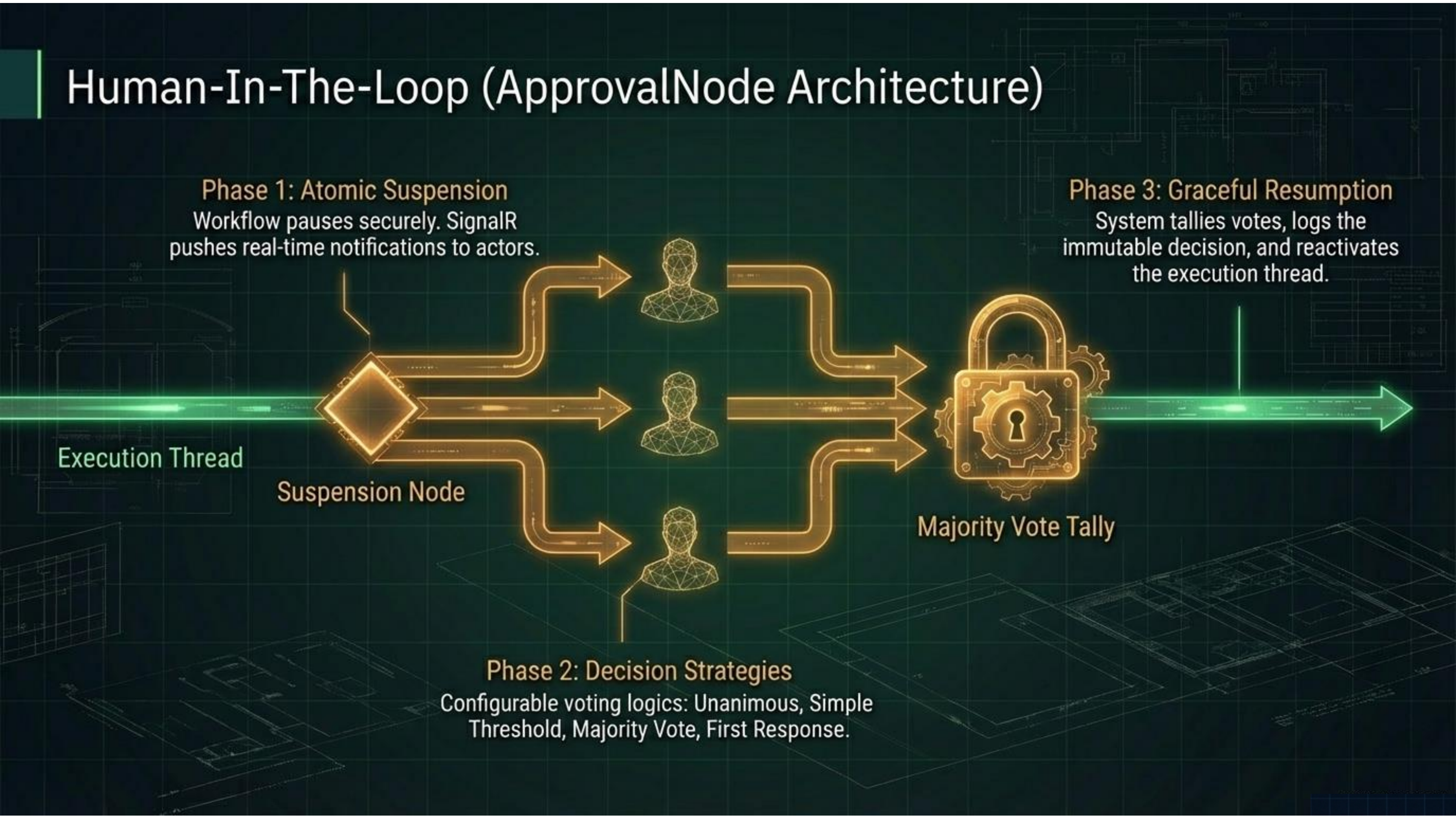
**Phase 3: Graceful Resumption**  
System tallies votes, logs the immutable decision, and reactivates the execution thread.

Execution Thread

Suspension Node

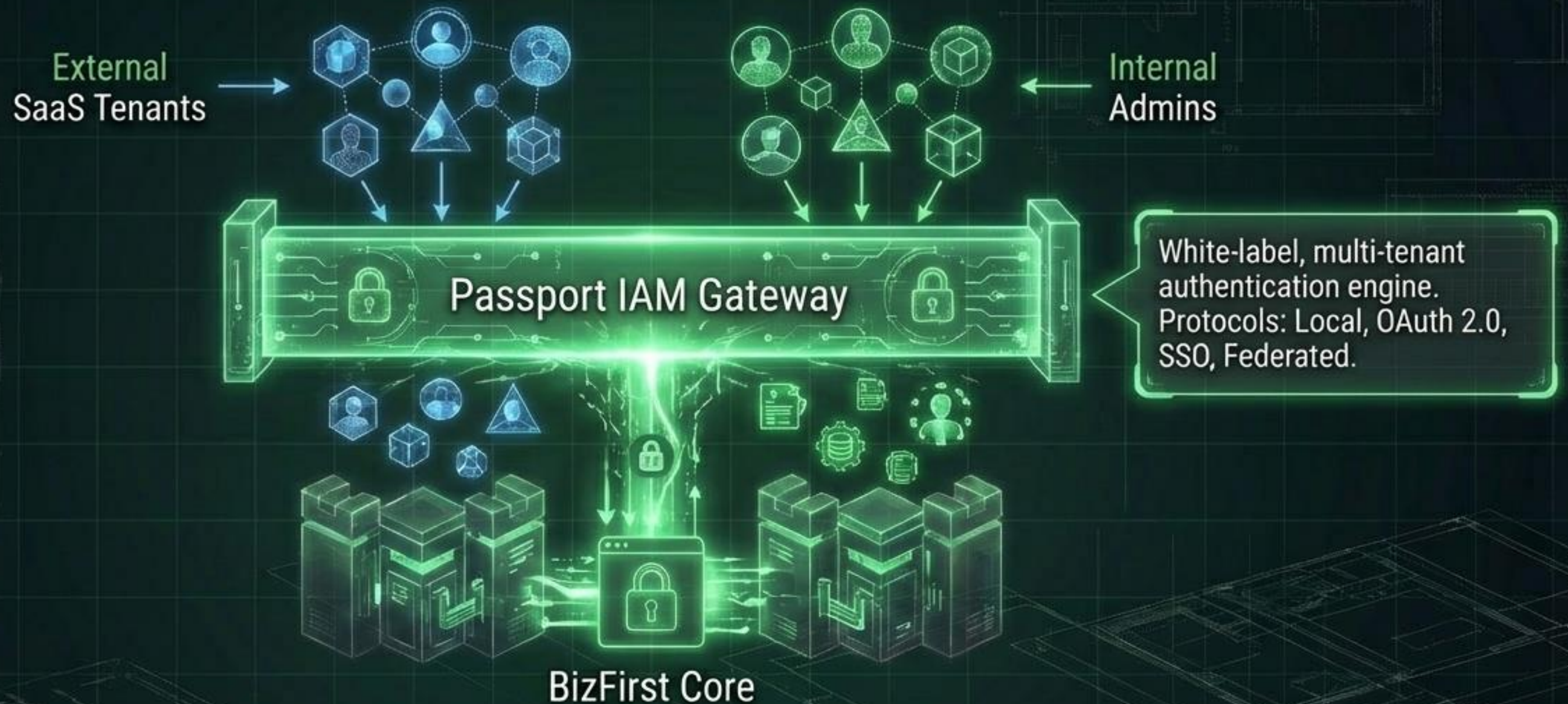
Majority Vote Tally

**Phase 2: Decision Strategies**  
Configurable voting logics: Unanimous, Simple Threshold, Majority Vote, First Response.





# Identity & Access: Passport IAM



**Value Proposition:** Securing the perimeter natively, eliminating the massive overhead, cost, and fragmented UX of external providers like Okta or Auth0.



# Enterprise Authorization: The 4-Layer Lock





# Data Foundation: DataOcean Architecture

## Soft-Delete & Temporal Filter

(CreatedOn, LastModifiedOn for enterprise compliance)

## Strict Tenant Isolation

Architectural mandate: TenantID exists in every single table, ensuring zero cross-tenant data leakage.



– **605+**

Normalized  
SQL Server Tables

– **48**

Distinct  
Data Domains

– **Multi-Tenant**

Enterprise Rigor



# Zero-Trust Isolation: Hardware TEE



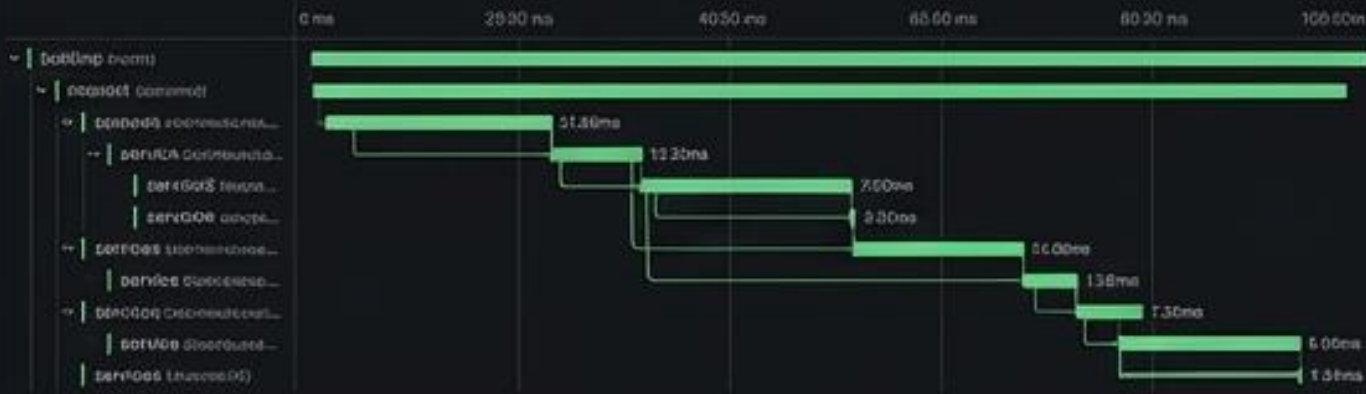
Hardware TEE (The Enclave)

- Hardware & Software Isolation (SGX/TDX)
- **The Zero-Trust Guarantee:** Sensitive operations (PCI-DSS, HIPAA) execute in enclaves where memory is mathematically inaccessible even to system operators.
- **Output:** Cryptographic proof of execution and deterministic output verification.



# Platform Observability (Telemetry & Tracing)

## Traces (Grafana Tempo)

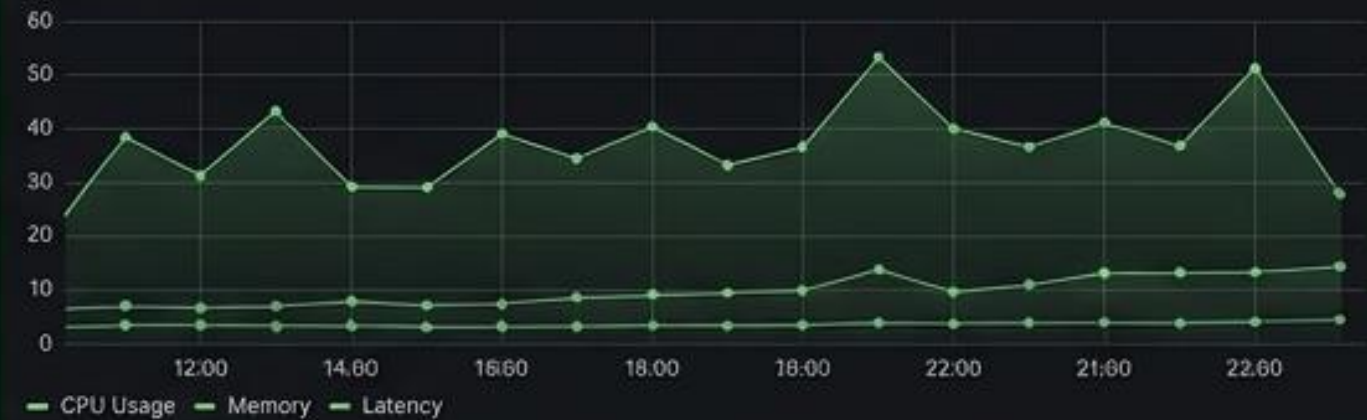


## Logs (Grafana Loki)

```
2023-09-07 12:12:13.315 INFO log acrdpendent: OpenTelemetry instrumentation, Zerv single A3peol/IDeTUTn56WnCRdCP105100L2M F06M400C%  
2023-09-07 15:10:05.385 INFO log acrdpendent: OpenTelemetry instrumentation, Zerv single A3peol/IDeTUTn56WnCRdCP105100L2M F06M400C%  
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```

## Metrics (Prometheus)

### System Health



### Business KPIs

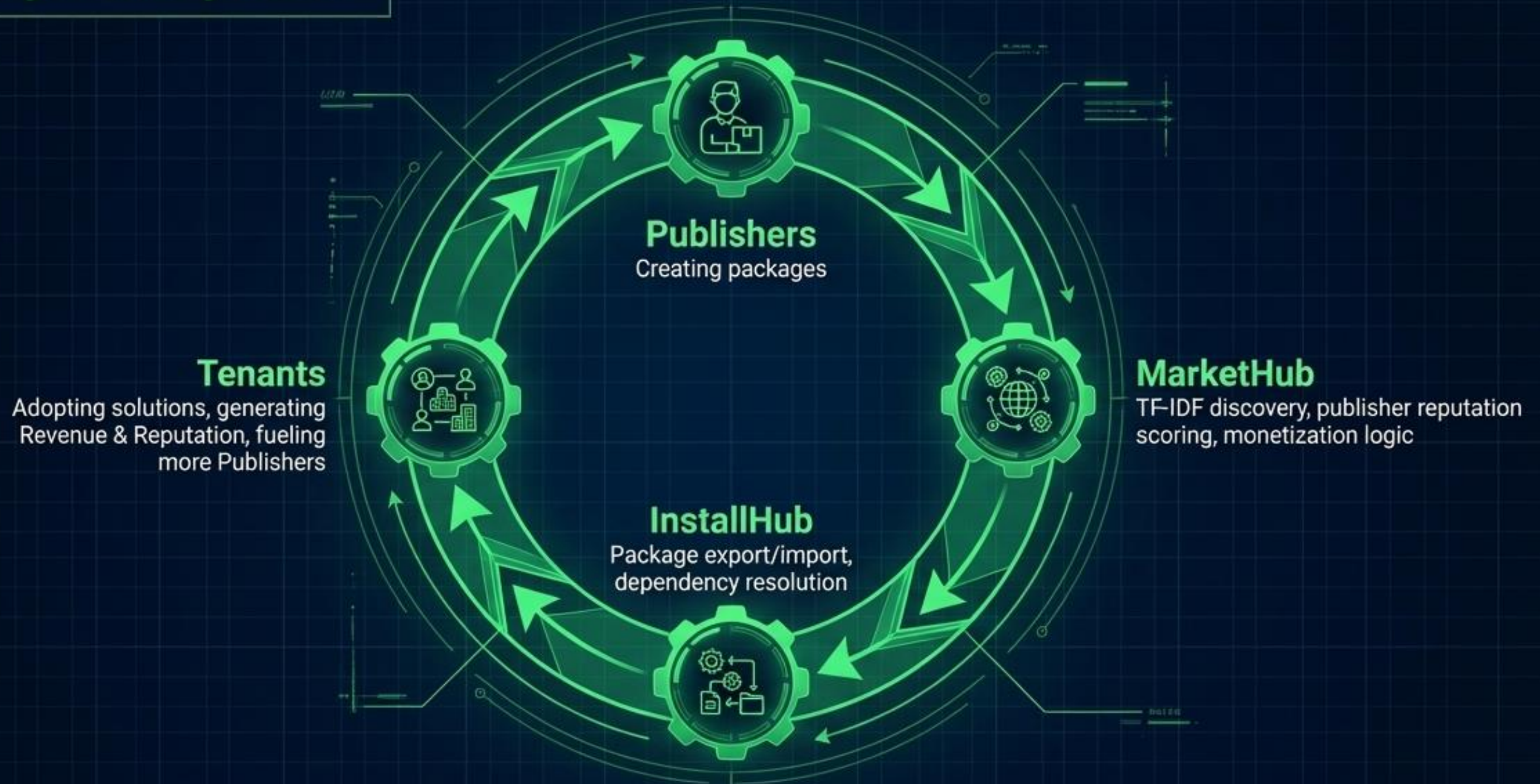


## The Telemetry Advantage

Zero-dependency OpenTelemetry instrumentation. Every single signal is automatically enriched with 'TenantID' and 'TraceID', allowing instantaneous cross-stack debugging from the API down to specific database queries, without third-party overlays.



# The Ecosystem Flywheel



The BizFirstAi Architecture isn't just a software stack;  
It is a **continuously compounding enterprise ecosystem.**